Multivariable calculus · Differentiation in two variables

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Partial derivatives, changes along a path, and a circle defined implicitly · compiled 17 October 2026

S1 Lesson: partial derivatives

S2 Lesson: total derivative

S3 Lesson: implicit curves

1 Partial derivatives and directional slopes

Definitions

Partial derivative. The derivative with respect to one coordinate while all other independent coordinates remain fixed. S1

Unit direction. A vector of length one. Normalise a supplied direction before using $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$. S1

Key points

1. Differentiate before substituting the evaluation point. Substitution too early can erase the variable that must change. S1

Why?

Why can the same surface have two coordinate slopes?
An x-section and a y-section follow different curves through the same point, so their slopes need not agree.
S1

Exam traps

Exam trap. The gradient is a vector. A directional slope is the dot product of that vector with a unit direction. S1

2 Total change and the chain rule

Definitions

Total differential. The linear part of the change in a differentiable function, $df = f_x\,dx + f_y\,dy$. S2

Composed function. The one-variable function obtained after replacing every changing input by its path expression. S2

Key points

1. A path can change several inputs at once; include one chain-rule term for each changing input. S2

Why?

Why does a differential differ from a finite change?
The differential keeps the linear part. Nonlinear terms remain in the exact finite increment.
S2

Exam traps

Exam trap. Do not write a finite-change approximation as an exact equality. State the base point and the increments. S2

3 Implicit curves and valid tangent formulas

Definitions

Implicit relation. An equation $F(x, y) = 0$ that describes admissible pairs (x, y). S3

Regular branch. A local representation of the curve as a differentiable function $y(x)$, justified here by $F_y \neq 0$. S3

Key points

1. Verify the point belongs to the relation and the chosen denominator is nonzero before evaluating an implicit derivative. S3

Why?

Why does a zero denominator require a new analysis?
It prevents division in the chosen formula. The curve may instead have a vertical tangent or fail to be regular; examine the defining equation.
S3

Exam traps

Exam trap. Do not evaluate a tangent at a point that is not on the curve. Do not turn division by zero into a slope of zero. S3

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